A. Additional evaluation results

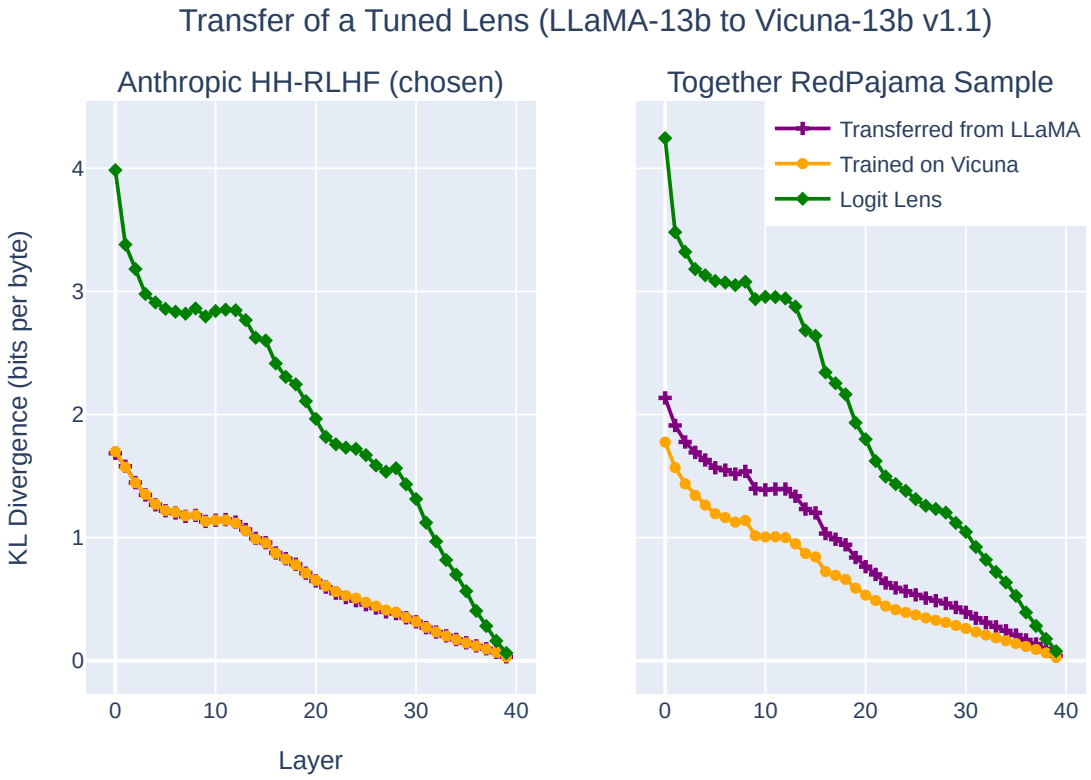

Figure 13. Comparing the performance of a lens specifically trained on Vicuna 13B vs a lens transferred from LLaMA 13B vs Vicuna 13B's Logit lens. Specifically, we messure the KL divergence between the lens at a specific layer and the model's final output.

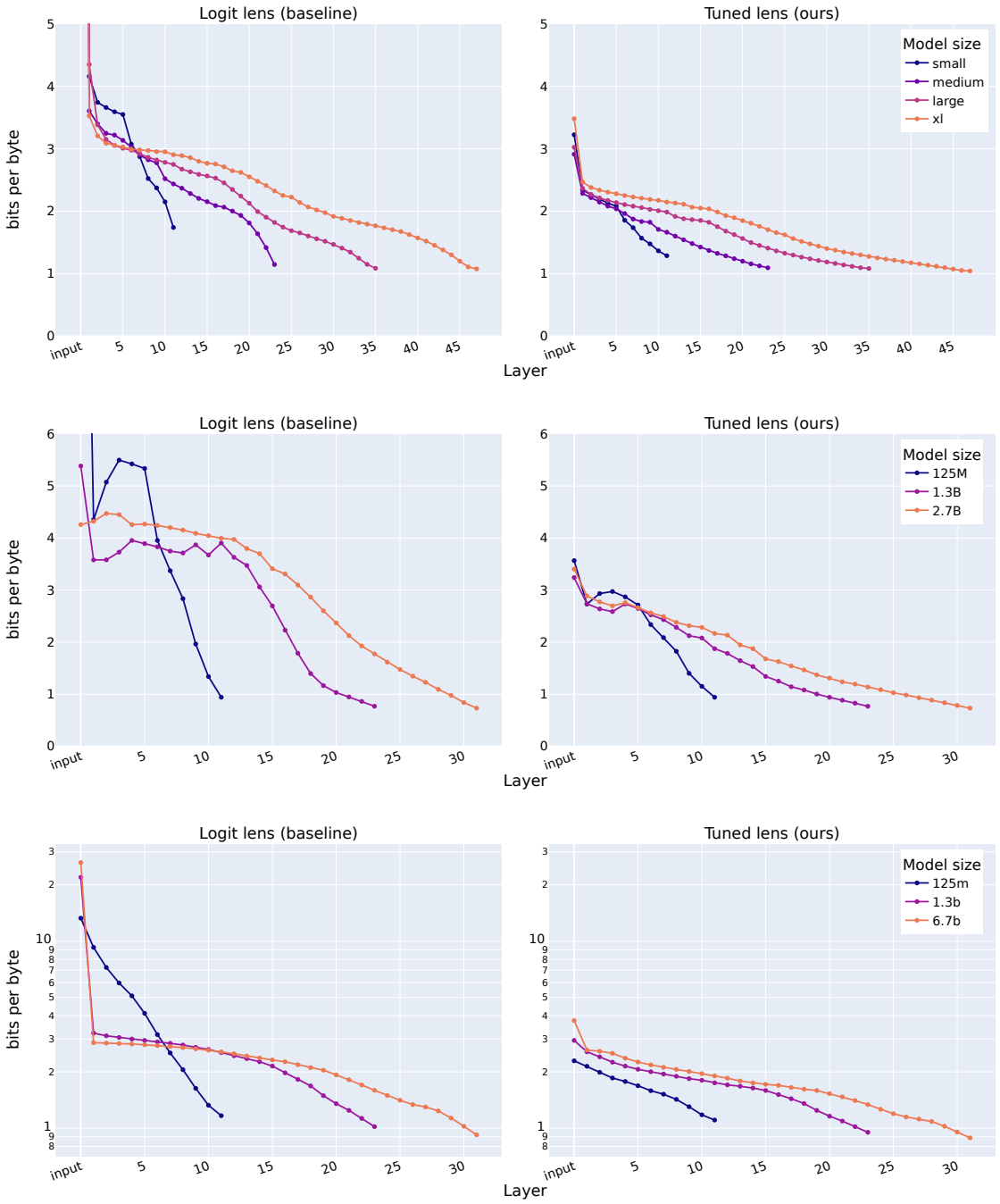

Figure 14. Perplexities of predictions elicited from OpenAI's GPT-2 models (top), EleutherAI's GPT-Neo models (middle), and Meta's OPT models (bottom). We omit OPT 350M as it uses a post-LN architecture.

B. Qualitative Results

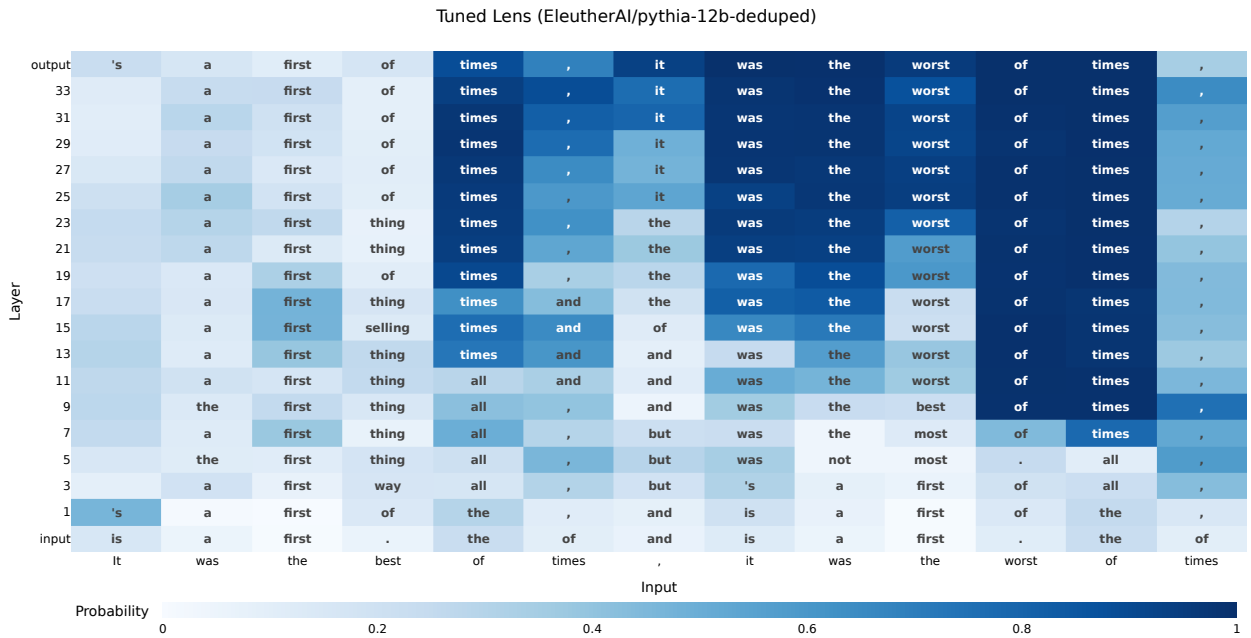

Figure 15. Tuned lens prediction trajectory for Pythia 12B on the first several words of Charles Dickens' "A Tale of Two Cities." Note that the latent prediction becomes very confident at early layers after the prefix "It was the best of times" has been processed, suggesting some degree of memorization.

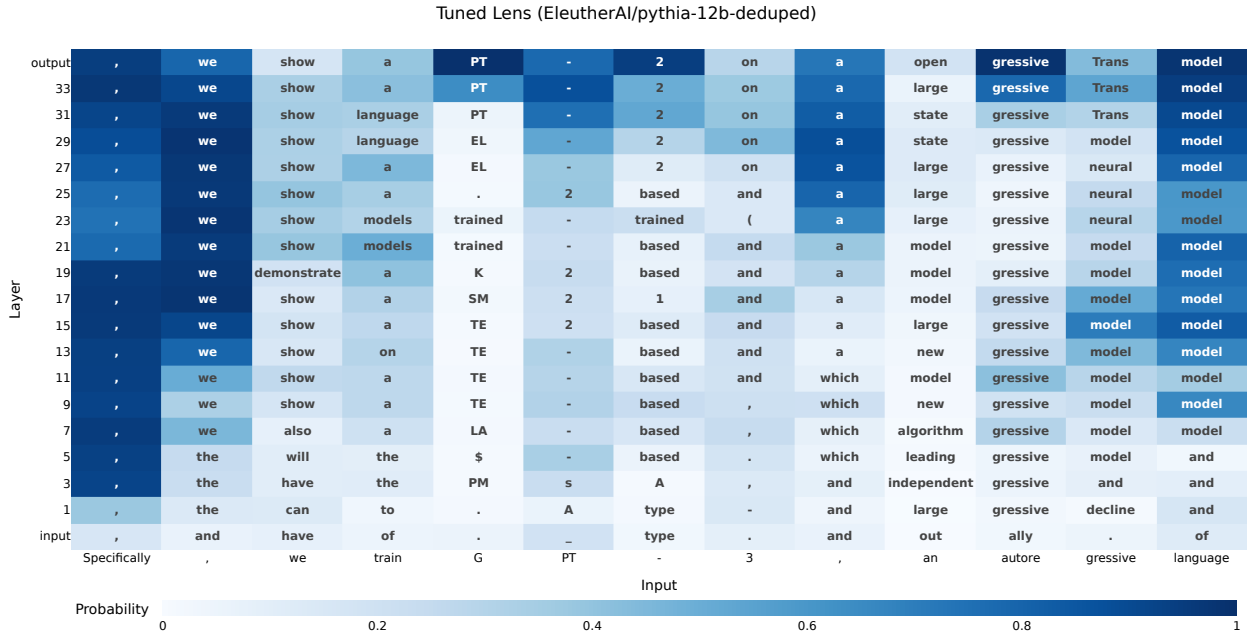

Figure 16. Tuned lens prediction trajectory for Pythia 12B prompted with the abstract of Brown et al. (2020).

B.1. Logit lens pathologies

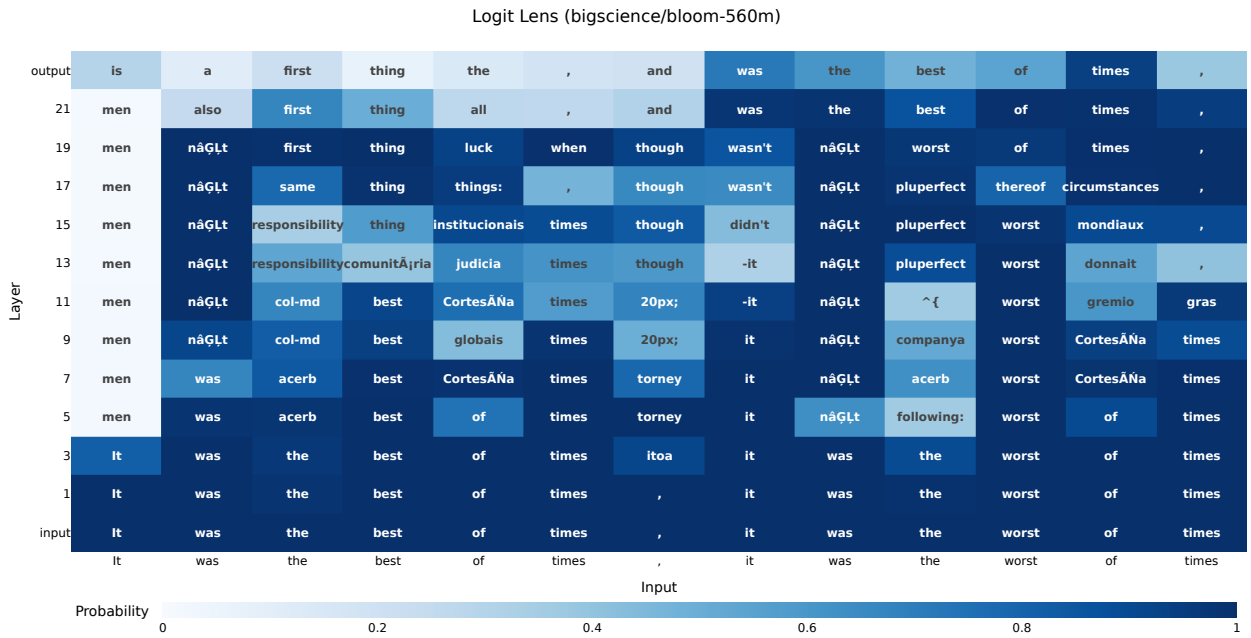

Figure 17. Logit lens prediction trajectory for BLOOM 560M. The logit lens assigns as very high probability to the *input* token at many layers and token positions, complicating the interpretation of the output.

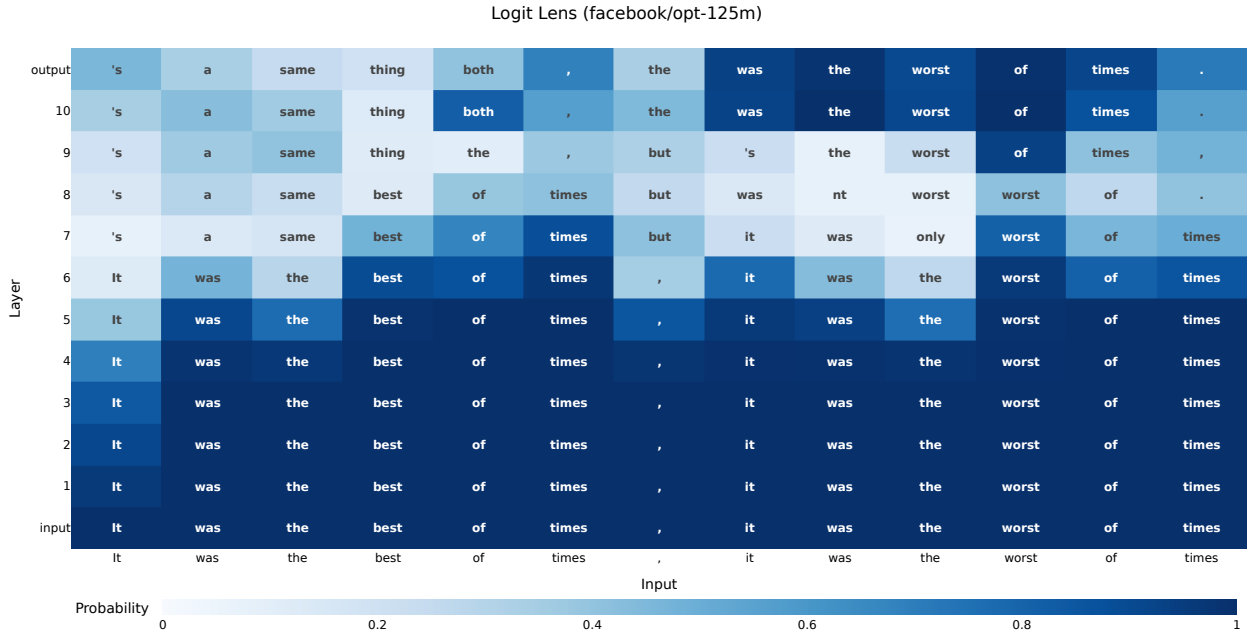

Figure 18. Logit lens prediction trajectory for OPT 125M, exhibiting a similar pathology to BLOOM 560M above.

C. Transformers Perform Iterative Inference

Jastrzębski et al. (2017) argue that skip connections encourage neural networks to perform "iterative inference" in the following sense: each layer reliably updates the hidden state in a direction of decreasing loss. While their analysis focused on ResNet image classifiers, their theoretical argument applies equally well to transformers, or any neural network featuring residual connections. We reproduce their theoretical analysis below.

A residual block applied to a representation $\boldsymbol{h}_i$ updates the representation as follows:

$$\boldsymbol{h}_{\ell+1} = \boldsymbol{h}_\ell + F_\ell(\boldsymbol{h}_\ell)$$

Let $\mathcal{L}$ denote the final linear classifier followed by a loss function. We can Taylor expand $\mathcal{L}(\boldsymbol{h}_L)$ around $\boldsymbol{h}_i$ to yield

$$\mathcal{L}(\boldsymbol{h}_L) = \mathcal{L}(\boldsymbol{h}_i) + \underbrace{\sum_{j=i}^{L} \left\langle F_j(\boldsymbol{h}_j), \frac{\partial \mathcal{L}(\boldsymbol{h}_j)}{\partial \boldsymbol{h}_j} \right\rangle}_{\text{gradient-residual alignment}} + \mathcal{O}(F_j^2(\boldsymbol{h}_j)) \tag{19}$$

Thus to a first-order approximation, the model is encouraged to minimize the inner product between the residual $F(\boldsymbol{h}_i)$ and the gradient $\frac{\partial \mathcal{L}(\boldsymbol{h}_i)}{\partial \boldsymbol{h}_i}$, which can be achieved by aligning it with the negative gradient.

To empirically measure the extent to which residuals do align with the negative gradient, Jastrzębski et al. (2017) compute the cosine similarity between $F(\boldsymbol{h}_i)$ and $\frac{\partial \mathcal{L}(\boldsymbol{h}_i)}{\partial \boldsymbol{h}_i}$ at each layer of a ResNet image classifier. They find that it is consistently negative, especially in the final stage of the network.

We reproduced their experiment using Pythia 6.9B, and report the results in Figure 19. We show that, for every layer in the network, the cosine similarity between the residual and the gradient is negative at least 95% of the time. While the magnitudes of the cosine similarities are relatively small in absolute terms, never exceeding 0.05, we show that they are much larger than would be expected of random vectors in this very high dimensional space. Specifically, we first sample 250 random Gaussian vectors of the same dimensionality as $\frac{\partial \mathcal{L}(\boldsymbol{h}_i)}{\partial \boldsymbol{h}_i}$, which is (hidden size) $\times$ (sequence length) $= 8{,}388{,}608$. We then compute pairwise cosine similarities between the vectors, and find the 5th percentile of this sample to be $\mathbf{-6 \times 10^{-4}}$. Virtually all of the gradient-residual pairs we observed had cosine similarities below this number.

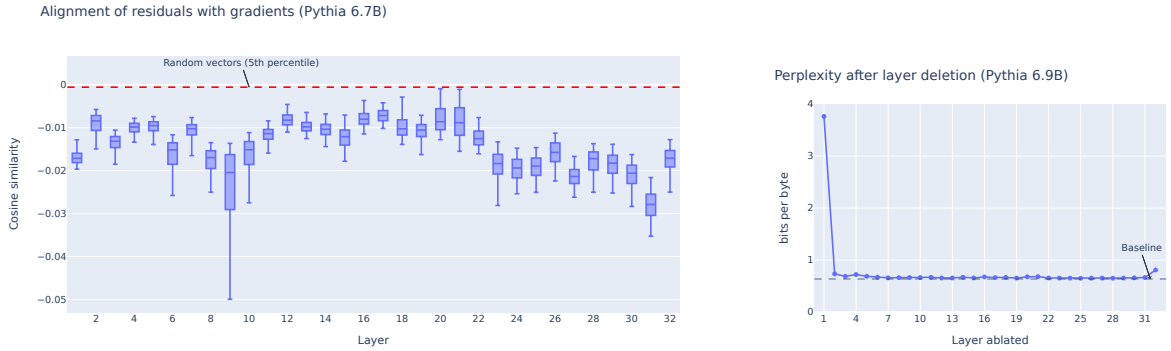

Figure 19. **Left:** Cosine similarity between the observed update to the hidden state and $\frac{\partial \mathcal{L}(\boldsymbol{h}_i)}{\partial \boldsymbol{h}_i}$. The similarity is almost always negative, and of much larger magnitude than would be expected by chance in this very high dimensional space. Boxplot whiskers are 5th and 95th percentiles. Results were computed over a sample of roughly 16M tokens from the Pile validation set. **Right:** Cross-entropy loss of Pythia 6.9B on the Pile validation set, after replacing each of its 32 layers with the identity transformation. The dashed line indicates Pythia's perplexity on this dataset with all layers intact.

C.1. Zero-shot robustness to layer deletion

Stochastic depth (Huang et al., 2016), also known as LayerDrop (Fan et al., 2019), is a regularization technique which randomly drops layers during training, and can be applied to both ResNets and transformers. Veit et al. (2016) show that